popsim: Reproducing Elements of Billion-Agent Social Simulators on a Single Machine, with a Behavioral-Adherence Benchmark for Persona Agents on Sentiment140

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Abstract

Population-scale social simulators such as Light Society (one billion agents) and MatrAIx (8.3 billion personas) require infrastructure most researchers cannot access. We show that their core methodology of human-grounded personas, tiered compute, and ground-truth validation runs on one commodity machine, using Sentiment140: 1.6M public tweets in under 300 MB of memory. We mine personas for the 6,245 users with at least 20 tweets each, and a mixture-of-models ladder cuts simulation cost 702.7× versus an all-LLM population. Validating on the real @-mention graph produces a negative result: DeGroot opinion diffusion ($r = 0.54$) underperforms per-user persistence ($r = 0.73$) at predicting future sentiment. Disposition beats diffusion. Sustained one-way (parasocial) attachments prove as numerous as strong mutual ties, motivating a near-zero-cost broadcast tier. We also release a fixed behavioral-adherence benchmark: per-user temporal 80/20 splits scored on 42,998 held-out tweets, where a zero-parameter persona baseline reaches 63.1% and MatrAIx's reported 91.5%, under a differing protocol, frames the gap. Code, artifacts, and an interactive explainer are public.

1 Introduction

Two recent systems define the frontier of population-scale social simulation, in a line that runs from generative agents [3] through million-agent open simulators such as OASIS [9]. Light Society [1] advances "earth-scale" simulation: over 10^9 agents whose states evolve through LLM-powered simulation operations, made tractable by a mixture-of-models engine combining full LLMs with distilled surrogates, and validated through trust-game and opinion-diffusion experiments. MatrAIx [2] contributes population realism of a different kind: 8.3 billion persona records spanning 1,290 attribute dimensions, with a curated million-persona subset grounded partly in real human profiles, and a headline evaluation showing persona agents express or appropriately suppress declared behaviors in 91.5% of trials.

Both systems presume compute, data pipelines, and engineering budgets beyond most academic and hobbyist settings. This paper asks a deliberately modest question: *which elements of this research program can be reproduced on one machine, over one public dataset, in an evening?* Our answers are embodied in **popsim**, built at a public build night, with all analysis running in-memory on a single node.[1]

Our contributions are:

1. **A single-machine, small-compute reproduction of elements of Light Society and MatrAIx**: human-grounded persona mining (§4); a mixture-of-models scale ladder with a measured 702.7× cost reduction (§5); and validation of simulated dynamics against held-out ground truth on a real interaction graph (§6), including a negative result that replicates, at the individual level and on a public corpus, the weak-contagion, strong-disposition pattern found in prior Twitter and Facebook studies [25][24][26].

2. **A new behavioral-adherence evaluation for persona agents on an existing public dataset** (§8): a fixed, reproducible protocol on Sentiment140 with a zero-parameter baseline of 63.1%, framed against MatrAIx's (protocol-distinct) 91.5% bar.

3. **An empirical account of relationship strength and parasocial structure** in the dataset's mention graph (§7), with a concrete

[1] Interactive explainer: `https://popsim.micahstubbs.ai`. Code and artifacts: `https://github.com/micahstubbs/pop-world-model-sim`.

architectural implication: population simulators need a broadcast tier, and it is nearly free.

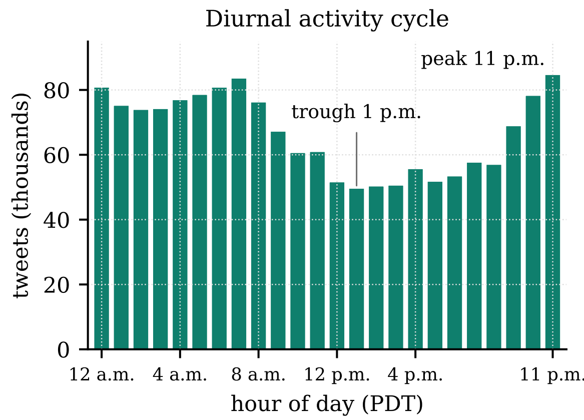

Figure 1: Diurnal activity cycle across the corpus (all 1.6M tweets).

2 Related Work

Population-scale simulation. Generative agent-based modeling descends from Park et al.'s generative agents [3], which demonstrated believable social behavior from memory-augmented LLM agents in a small sandbox town. Open simulators have since pushed scale: OASIS [9] models X- and Reddit-style platforms with up to 10^6 agents, AgentSociety [10] simulates tens of thousands of agents in a data-grounded urban environment, Y Social [11] provides an open social-media digital twin, and SocioVerse [12] frames simulation as a world model aligned to a pool of ten million real users. Light Society [1] formalizes society simulation as structured transitions of agent and environment states governed by LLM-powered operations, scaling to 10^9 agents grounded in World Values Survey profiles via a mixture-of-models engine that routes requests among full LLMs, distilled surrogates, and precomputed lookup tables; its successor APS [13] reports a measured 381× reduction in LLM calls at 10^7 agents. MatrAIx [2] builds an 8.3B-record persona database (599,847 human-grounded profiles among the curated subset) and evaluates behavioral adherence across four environments. Closest in spirit to our framing, Itkin [14] shows that LLM-agent societies can be run on a laptop by cloning each agent's policy into a few-parameter surrogate; we pursue the complementary question of which elements survive when the population, the interaction graph, and the validation target are all real.

Tiered compute. Heterogeneous agent cost is now a recurring design: core LLM agents with diffusion-model agents for the remaining population [15], topology-aware grouping of agents into shared backbone units [16], and Light Society's surrogate distillation [1]. Ziems et al. [17] find that model scale helps opinion and behavior modeling for well-represented populations but scales slowly for longitudinal forecasting, which motivates spending frontier compute only on users with dense signal.

Evaluating persona agents. Park et al. [18] ground agents in two-hour interviews with 1,052 people and score held-out survey responses against each participant's own test–retest consistency; Twin-2K-500 [19] supplies a survey-based dataset with a repeat wave for the same purpose. On social media, BluePrint [20] clusters anonymized Bluesky users into personas and poses next-action prediction; BehaviorChain [21], TwinVoice, and PersonaArena [22] test persona simulation with constructed or judge-scored scenarios. Wu et al. [23] review this literature and find that fewer than half of validation studies report behavioral variance, which is why we report the per-user adherence distribution alongside the aggregate. Our benchmark differs from all of these in scoring individual (not clustered) users on real held-out behavior from a fully public corpus.

Sentiment dynamics and contagion on Twitter. Whether sentiment spreads over ties has a long empirical record. Kramer et al. [24] established causal emotional contagion on Facebook with very small effect sizes; Ferrara and Yang [25] measured about a 4% over-exposure effect on Twitter with a scarcely-susceptible majority; Charlton et al. [26] found that sentiment levels of stable @-mention communities persist over months, with shifts traceable to external events. Aral et al. [27] and Shalizi and Thomas [28] show that observational contagion estimates are confounded with homophily, while Tan et al. [29] show that @-mention neighbors are nonetheless a useful static prior for user-level sentiment. Our diffusion experiment (§6) quantifies this trade-off predictively on a public corpus: DeGroot averaging [8] over the real mention graph versus each user's own history, with the Friedkin–Johnsen anchored variant [30] as the natural intermediate left to future work.

Relationship structure. Kwak et al. [6] established the large-scale structural view of Twitter; Huberman et al. [31] showed that the sparse network of reciprocal @-interactions, not the follower graph, is the network that matters; Cha et al. [32] showed that mention-based attention concentrates on celebrities independently of follower count. Horton and Wohl [7] introduced the parasocial-interaction construct we operationalize on mention pairs.

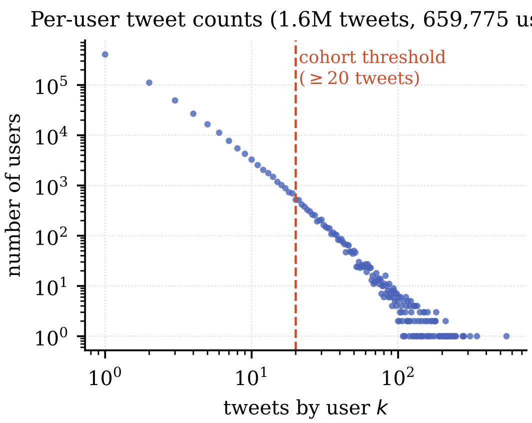

Figure 2: Per-user tweet counts are heavy-tailed. The cohort threshold ($\geq$ 20 tweets) admits 6,245 users covering 226,202 tweets.

Data. Sentiment140 [4] provides 1.6M tweets from April–June 2009, distantly labeled for sentiment by emoticons. It remains one of the few large fully-public full-text tweet corpora: since the 2023 restriction of the X API, tweet-ID "hydration" workflows that most academic Twitter datasets rely on are effectively defunct, making full-text corpora the practical substrate for new work. To our knowledge it has been used almost exclusively for classifier benchmarking; we use its per-user timelines and @-mention graph instead. TweetEval [5] standardizes tweet classification benchmarks and supplies pretrained classifiers usable as realism scorecards; Golder and Macy [33] document the diurnal mood cycle we recover in Figure 1.

3 Dataset and Cohort

Sentiment140 loads from one CSV into 295 MB of memory: 1,600,000 tweets, 659,775 distinct users, 48 days, balanced 800K/800K negative/positive labels. Figure 2 shows the heavy-tailed per-user activity distribution. We define the **persona cohort** as the 6,245 users with at least 20 tweets (226,202 tweets; mean 36.2, median 28 per user; mean per-user activity

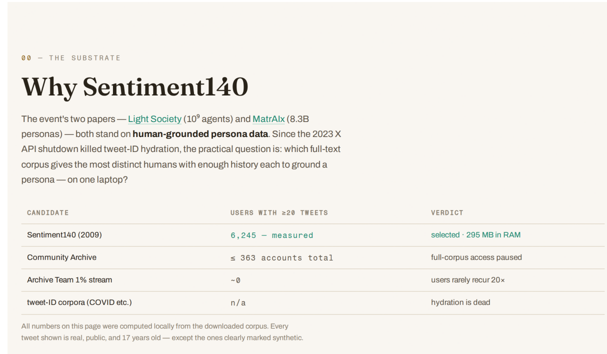

Figure 3: Dataset selection in the interactive explainer (`https://popsim.micahstubbs.ai/#data`): candidate corpora compared on the number of distinct users with at least 20 tweets.

span ~50 days). The corpus retains a strong diurnal rhythm (Figure 1), with a peak at 11 p.m. and a trough at 1 p.m. PDT, usable for calibrating agent posting schedules [33]. Selection among candidate corpora was itself empirical (Figure 3): the Community Archive holds only 363 accounts; the Archive Team 1% stream rarely observes a user 20 times; ID-only corpora cannot be hydrated. Sentiment140 maximizes the number of distinct humans with enough history to ground a persona while fitting in RAM.

4 Persona Mining

Following MatrAIx's human-grounded persona construction at hackathon scale, each cohort user becomes a persona card computed from their real timeline: sentiment disposition (share of positive tweets), verbosity (words/tweet), mention rate, peak posting hour, activity span, and characteristic vocabulary. Unlike survey-grounded profiles, these personas carry temporal behavior (posting cadence, diurnal phase) directly usable to drive simulation ticks. The cohort skews positive (59% vs. the corpus's balanced 50%), a selection effect worth modeling rather than ignoring: highly active users differ from the population they are drawn from.

5 The Mixture-of-Models Scale Ladder

Heterogeneous agent cost is Light Society's central efficiency device [1] and has since appeared as core/background splits [15], structural grouping [16], prototype routing [13], and whole-agent surrogates [14]: full LLMs only where fidelity demands them.

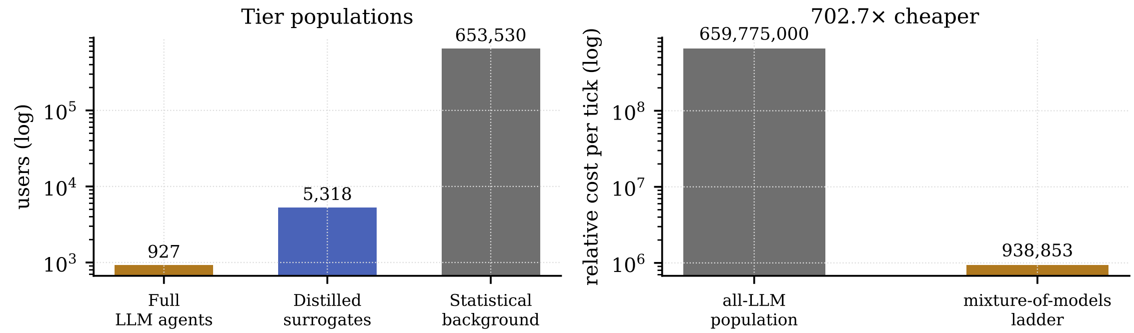

Figure 4: The scale ladder. Left: tier populations (log scale). Right: relative cost of one simulation tick, all-LLM versus the ladder.

We instantiate the same ladder sized to one node (Figure 4): **full LLM agents** for the 927 users with $\geq$ 50 tweets (enough signal to condition a frontier model); **distilled surrogates** (persona-conditioned small models) for the 5,318 users with 20–49 tweets; and a **statistical background** tier (rate and sentiment samplers) for the remaining 653,530 users. Under a relative cost model of $1000 : 1 : 0.01$ units per agent-tick, the ladder costs 9.39×10^5 units per tick versus 6.60×10^8 for an all-LLM population, a 702.7× reduction under the nominal cost model, with the full-LLM tier reserved for the users who carry most of the observable signal. APS [13] reports a measured 381× reduction at 10^7 agents against a full-LLM reference; we do not measure fidelity loss here.

6 Does Sentiment Diffuse? A Validated Negative Result

The cohort contains a real directed interaction graph: 31,515 @-mention events between cohort members.

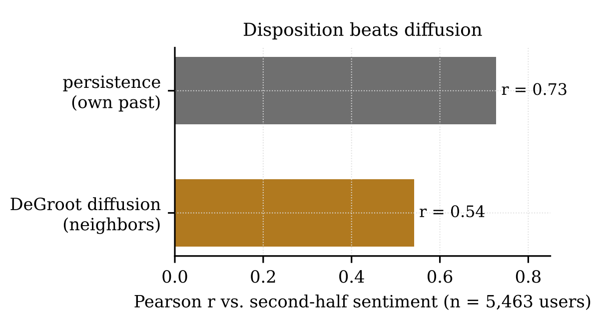

Figure 5: Predicting second-half sentiment from first-half state. Simple persistence outperforms DeGroot diffusion over the real mention graph.

Light Society validates simulations with opinion-diffusion experiments; we run the corresponding test *against ground truth.* Each user is seeded with their first-half sentiment; DeGroot mixing [8] ($s_{t+1} = 0.7\, s_t + 0.3\, \bar{s}_{\mathcal{N}}$, 10 steps, edge weights = mention counts) evolves states over the real graph; we then ask which predictor better recovers each user's *second-half* sentiment (users with $\geq$ 5 tweets in each half; $n =$ 5,463).

Persistence wins (Figure 5): $r = 0.73$ for a user's own past versus $r = 0.54$ after diffusion; restricting to graph-connected users widens the gap (0.72 vs. 0.40). Neighbor-mixing strictly destroys predictive information at this horizon. **Disposition beats diffusion**: over 48 days, who a person is outweighs who they talk to. This is the predictive face of a well-known identification problem: in observational network data, apparent contagion is largely homophily [27], and the two are generically confounded [28]. Our test does not show that influence is absent, only that averaging over neighbors adds no information beyond a user's own history at a 24-day horizon, in line with the ∼4% exposure effects Ferrara and Yang [25] measure and the month-scale stability Charlton et al. [26] observe on the mention graph. For simulator design the implication is direct: a population world model that over-weights social contagion will be *less* accurate than one grounded in stable personas, which is why persona grounding (§4) matters. Static neighbor information may still help as a prior [29]; what fails is dynamic mixing.

7 Relationship Strength and Parasocial Structure

To model whom agents should talk to, we quantify relationship structure over the full mention graph

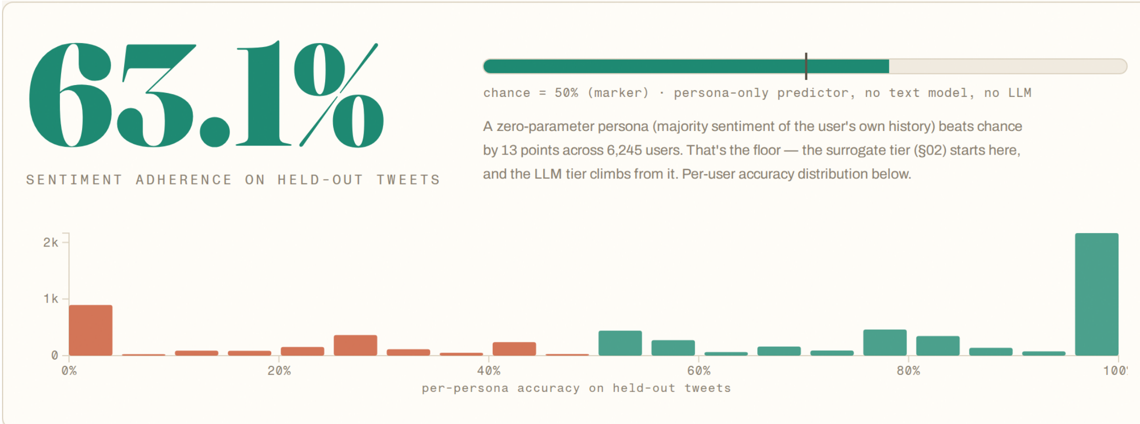

Figure 6: The benchmark section of the interactive explainer (`https://popsim.micahstubbs.ai/#adherence`): the 63.1% zero-parameter baseline against the 50% chance marker, and the per-persona accuracy distribution over held-out tweets.

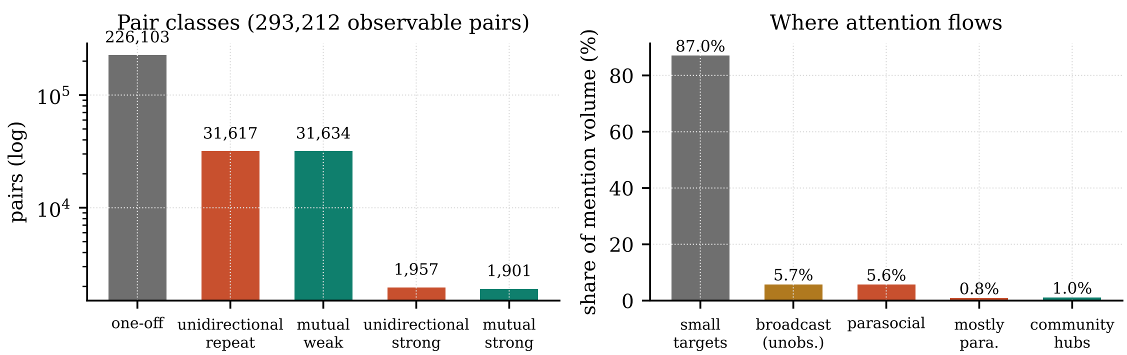

Figure 7: Left: relationship classes among the 293,212 pairs with observable reciprocity. Right: share of total mention volume by target class; parasocial + broadcast attention outweighs reciprocating community hubs by an order of magnitude.

(all 1.6M tweets). For unordered pairs where both users author tweets in the sample (so silence is observed, not censored; n = 293,212 pairs), we score reciprocity $r = \min(w_{ab}, w_{ba}) / \max(w_{ab}, w_{ba})$ and strength $\log(1+\text{total})\cdot\log(1+\text{days})\cdot(1+r)$, and classify pairs by mutuality and volume (Figure 7, left). Genuine relationships are rare: 77% of pairs are one-off contacts, echoing Huberman et al.'s finding that the network of reciprocal @-interaction is far sparser than the declared graph [31]. Sustained one-way attachments (*unidirectional strong*: ≥ 5 mentions, zero returned; 1,957 pairs) are as numerous as strong mutual bonds (1,901 pairs). For every observable friendship there is roughly one unrequited attachment.

At the account level, among in-sample targets with ≥ 20 distinct mentioners, 408 reciprocate under 5% of their audience, operationally *parasocial* [7], including 2009-era celebrities (`wossy`, `ashleytisdale`, `tomfelton`) who tweet but do not talk back (the mention concentration on celebrities that Cha et al. [32] documented), while 163 *community hubs* of comparable audience reciprocate 30–53%. A further 524 broadcast targets (e.g. `jonasbrothers`: 1,901 distinct fans in 48 days) never author a sampled tweet, so their reciprocity is unobservable and we label rather than classify them. By volume, 12.1% of all mention traffic flows to broadcast/parasocial accounts versus 1.0% to reciprocating hubs (Figure 7, right).

The architectural implication joins §5: simulated populations need a **broadcast tier**, and it is nearly free, since parasocial edges are one-directional by construction. The celebrity side requires no agent at all, only a content feed.

8 A Behavioral-Adherence Benchmark on Sentiment140

MatrAIx's headline metric, whether persona agents behave like their personas, currently lacks a public, fixed-protocol instantiation on real held-out behavior of individual users: Park et al. [18] and Twin-2K-500 [19] score held-out survey items, and BluePrint [20] scores next actions for anonymized persona clusters rather than individuals. We contribute one on an existing fully-public corpus, with live results in the interactive explainer (Figure 6).

Protocol (fixed). Cohort: the 6,245 Sentiment140 users with $\geq$ 20 tweets. Per user, order tweets by time; train = first 80%, test = last 20% (minimum 4). Metric: fraction of the 42,998 held-out tweets whose sentiment label the persona model predicts, reported overall and as the per-user distribution. Rules: no access to any user's test tweets at training or persona-construction time; train-tweet text is unrestricted (features, pretrained classifiers, persona-conditioned LLM prompting, fine-tuning).

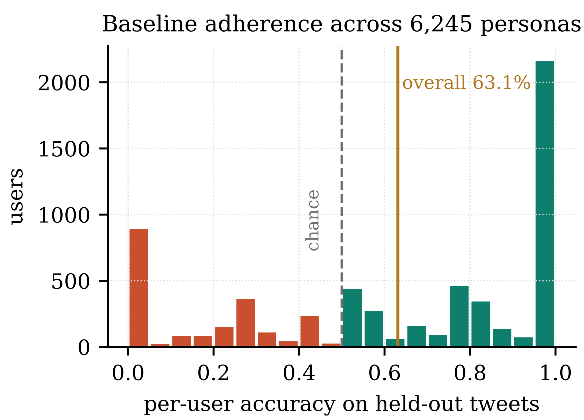

Figure 8: Per-user adherence of the zero-parameter persona baseline on held-out tweets. Overall: 63.1% over 42,998 tweets.

Baseline. The zero-parameter persona, which predicts each user's train-majority sentiment, attains **63.1%** (chance: 50%, label-balanced corpus). Figure 8 shows the per-user distribution: bimodal, with a large mass of highly consistent users and a long tail of volatile ones for whom persona-only prediction fails. This is the floor a surrogate tier starts from and an LLM tier must climb. Reporting the distribution, not only the mean, follows Wu et al.'s [23] recommendation that simulation evaluations report behavioral variance.

The bar. MatrAIx reports 91.5% behavioral adherence across ten attributes in four environments. The protocols are not comparable: theirs measures declared-attribute expression in synthetic interactions, while ours measures real held-out behavior. We therefore frame 91.5% as an aspirational bar on our protocol, not a record to break on equal terms. The benchmark poses an open question: how far above 63.1% can persona-conditioned models go on real, noisy, distantly-labeled human behavior?

9 Limitations

Our substrate is 2009 Twitter: 140-character posts, no quote tweets, no algorithmic feed; behavioral constants observed here need not transfer to modern platforms. Sentiment labels are distant (emoticon-derived) and noisy. The corpus is a partial sample, which censors reciprocity; we mitigate by restricting reciprocity claims to in-sample authors with audience $\geq$ 20 and labeling out-of-sample targets as unobserved. The scale-ladder cost model uses nominal relative unit costs, not measured wall-clock inference costs. The diffusion experiment tests one model (DeGroot, $\alpha = 0.7$, 10 steps) at one horizon; richer contagion models, most naturally Friedkin–Johnsen [30], which anchors each agent to its initial state and interpolates between persistence and DeGroot, could yet beat persistence; and because homophily and influence are not separable in observational data [28], our result concerns prediction, not causation. Finally, the adherence benchmark measures a single behavioral dimension (sentiment); MatrAIx-style multi-attribute adherence on this cohort is natural future work.

10 Conclusion

The methodology of billion-agent social simulators is not confined to billion-agent budgets, a point made independently by Itkin [14] for LLM-cloned surrogates; our contribution is to do it with real people, a real graph, and real held-out behavior. On one machine and one public dataset we reproduced the persona-grounding and mixture-of-models patterns of Light Society and MatrAIx, validated a diffusion assumption against ground truth (and found it wanting), mapped the parasocial structure a realistic simulator must include, and left behind a fixed, public behavioral-adherence benchmark with a 63.1% baseline and a 91.5% bar. The gap between those numbers is, we hope, an invitation.

Artifact Availability

The interactive explainer shown in Figures 3 and 6 is live at `https://popsim.micahstubbs.ai`. All analysis code, data pipelines, the benchmark protocol, and the source of this paper are at `https://github.com/micahstubbs/pop-world-model-sim`; the deployed site source is at `https://github.com/micahstubbs/popsim`.

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